MIL-A-36494
7 November 1966

MILITARY SPECIFICATION

AUDIOMETER, PORTABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of lightweight, portable, single channel, manually operated audiometer, capable of performing screening examinations and basic threshold evaluations with masking noise and bone conduction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-P-116	Preservation, Methods of

FSC 6515

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION, INC. (ASA)

ASA C73.11-1963 Dimensions of Caps and Receptacles: 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type, General Purpose.

(Application for copies should be addressed to American Standards Association, Inc., 10 East 40th Street, New York, N.Y. 10016.)

HEARING AID INDUSTRY CONFERENCE (HAIC)

Interim Bone Conduction, Thresholds for Audiometry.

(Application for copies should be addressed to Hearing Aid Industry Conference, 375 Valley Brook Road, Canonsburg, Penna. 15317.)

INTERNATIONAL STANDARD ORGANIZATION (I.S.O.)

R389-1964 Standard Reference Zero for the Calibration of Pure-Tone Audiometers.

(Application for copies should be addressed to American Standards Association, Inc. 10 East 40th Street, New York, N.Y. 10016.)

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NATIONAL BUREAU OF STANDARDS (NBS)

Close Coupler Type 9A.

(Application for copies should be addressed to Superintendent of Documents, Washington 25, D.C.)

UNDERWRITERS LABORATORIES, INC. (UL)

(Application for copies should be addressed to Underwriters Laboratories, Inc. 207 East Ohio Street, Chicago, Ill. 60611.)

3. REQUIREMENTS

3.1 First Article. Unless otherwise specified, prior to the start of production of the contract quantity, sample(s) of the audiometer shall be assembled, completely finished, and tested by the contractor. The sample(s) shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first article(s), acquisition of materials, components, or commencement of production is at the sole risk of the contractor. Accepted sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit to the contracting agency, prior to delivery, proof that the equipment he will supply under this specification conforms to the applicable requirements of Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the equipment conforms to this requirement.

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3.2.2 In lieu of the label or listing, the contractor may submit, prior to delivery, independent proof satisfactory to the contracting agency that his equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

3.2.3 Compliance with Underwriters' Laboratories, Inc. requirement regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.3 Material. Material shall be of the highest quality throughout, durable, and suitable in all respects for the purpose intended.

3.4 Construction. The audiometer shall be contained in a rugged, shock-resisting, plastic covered, wood cabinet. The cabinet shall be provided with a corrosion-resisting metal control panel and protecting covers with a protectively lined storage compartment. A suitable handle shall be attached to the cabinet for carrying. A fuse holder and slow-blow type fuse shall be provided as a precaution against overloading. Also provided on the cabinet shall be the necessary patch-cord connections to extend receivers from jacks on the audiometer to the audiometric test booth, to enable two room operation.

3.4.1 Control panel. The control panel shall be located on the front of the cabinet and shall be provided with at least the following: power ON-OFF, and tone interrupter switches, a variable attenuator of 5 decibels steps or less, a self-aligning frequency selector and a masking control.

3.4.2 Storage compartment. The storage compartment shall contain mountings for securing six replacement fuses and shall be capable of housing and protecting all accessories included in the unit.

3.4.3 Accessories. The following items shall be included in the unit of issue for the audiometer:

Two air conduction receivers with cord and head band.

A bone conduction receiver with cord and head band.

Patch cord set (for adapting unit for use with FSN 6515-065-9600 Booth, Audiometeric Examination).

Six replacement fuses (slow-blow type).

One pr. extra air receiver cord.

A two prong adapter plug with grounding pigtail.

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3.4.4 Receivers. The earphone cups shall be fabricated of insulated type plastic. Each cup shall hold layers of sound insulating material and shall be provided with a large contour, non-liquid, ear pad, manufactured of material which does not take a permanent set. The pad shall completely enclose the ear when used. Each cup shall enclose a dynamic moving coil type speaker.

3.5 Design. The audiometer shall be a light weight, portable unit, designed to perform screening examinations and basic threshold evaluations. The audiometer shall have a uniform zero reference level and shall be capable of performing the following tests, without the use of accessories not currently included as component parts of the unit:

(a) Pure tone air conduction

(b) Pure tone air conduction with masking

(c) Pure tone bone conduction

(d) Pure tone bone conduction with masking

3.5.1.1 Attenuation. The audiometer shall be calibrated with five decibels steps or less. The accuracy of the attenuator shall be within plus or minus 1.5 decibels per five decibels steps and plus or minus 2-1/2 decibels cumulative over the entire intensity range. The range of attenuation shall be from 0 to +110 decibels (see 3.7).

3.5.1.2 Wave form. The purity of the tone in the air conduction receiver shall be such that the harmonics at any particular frequency shall be at a level not less than 25 decibels below the fundamental tone as measured by acoustic output from receiver with a close coupler National Bureau of Standards (NBS) type 9A. Accessory noises such as line hum, click of interrupting switches, shall not in any way interfere with the test tone audible at intensity settings as high as 85 decibels.

3.5.2 Frequency. The audiometer shall generate test tones of at least the following identified fixed frequencies with a 3% accuracy: 125, 250, 500, 1000, 2000, 3000, 4000, 6000 and 8000 cycles per second (cps) for air conduction measurements and 250, 500, 1000, 2000, 3000 and 4000 cps for bone conduction measurements. The frequency selector shall be of the self-aligning type and shall be supplied with a large dial numerically marked to indicate at least the test frequencies listed above.

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3.5.3 Air conduction receivers. The test receiver shall be provided with the left and right receivers of different colors or distinctly marked for rapid identification. The two units shall be joined by a suitable adjustable headband that exerts a pressure of 1000 to 2000 grams laterally. The test receivers shall provide relative purity of sound production from 50° to 120° F with a maximum out put between 67DB to 120 DB.

3.5.4 Bone conduction. The audiometer unit shall be provided with a small lightweight, hearing aid type bone conduction receiver of not less than 1 square centimeter in size, designed to minimize air conduction radiation so that radiated sound reaching the tympanum through the auditory meatus shall not influence the validity of the bone conduction measurement. The conductor and headband shall be so designed that when in use the static force of the conductor shall be adequate to prevent the dynamic force from effecting a physical beat of the conductor against the area under test. The bone conduction receiver shall be calibrated in accordance with the Hearing Aid Industrial Conference (HAIC) Interim Bone Conduction Thresholds for Audiometry.

3.5.5 Tone interrupter. The tone interrupting device shall be a lever type switch mounted on the control panel and enabling reception of tones which are normally ON-manually OFF, or normally OFF--manually ON. The operation of the switch shall produce no intensity overshoot and no extraneous or transient frequencies.

3.5.6 Masking device. The unit shall be provided with a masking device to prevent carrying over of sound to the better ear when making bone and air conduction tests. The selector and control for the masking sound shall be calibrated for effective masking with a range of at least 20 to 60DB, but not to exceed 120 DB SPL and variable steps of 10DB. The control, which regulates the intensity of the masking tone shall indicate steps of not more than 10DB. When used, the masking tone shall be automatically transmitted to the ear opposite the one under test.

3.5.7 Calibration. The audiometer shall be calibrated according to the International Standards Organization (ISO) Publication 177 of 1965.

3.5.8 Electric power source and cord. The audiometer shall be designed to operate on a power source of 105-125 volts, or 210-250 volts (see 6.1), 50-60 cycle A.C. The audiometer shall have an internal voltage regulation circuit so designed that when the line voltage varies from 105 to 125 volts, or from 210 to 250 volts, the acoustic output level at 60 decibels shall not vary more than ± 1 decibel from its value at the nominal line voltage of 115 or 230 volts. The electric power cord shall be 3-wire, grounding type, having a minimum length of six feet. The power cord shall be furnished with a three prong plug for use with receptacles specified in A.S.A C73.11-1963 for Caps and Receptacles, 125 volts, 15 amperes, 2-pole, 3-wire, Grounding Type. A two prong adapter plug with grounding pigtail for use with two conductor electrical outlets shall be provided with each unit.

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3.6 Identification marking. Each audiometer shall be permanently and legibly marked with at least the following information:

Federal Stock Number

Manufacturers name or registered trademark

Model and serial number

The letters: U. S.

Electrical ratings (volts, amperes, watts)

Markings secured by adhesive shall not be used.

3.7 Audiogram charts. - One hundred audiogram charts suitable for graphing results of examination shall be furnised with each audiometer.

3.8 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. This information shall specifically include instructions, drawings, and a list of high mortality parts. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment (see 6.1). The preliminary service data shall be submitted sufficiently early to permit adequate review (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope, and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located. In addition the contractor

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shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.9 Workmanship. Workmanship shall be first class throughout. The finished audiometer shall be free of defects which detract from the appearance or impair the serviceability of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier shall utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL(defects per hundred units)
For visual examination		
Major defects	II	1.0
Total (major and minor combined)	II	4.0

4.2.1 For tests. Sampling for tests shall be conducted in accordance with Inspection Level S-1 of MIL-STD-105. The acceptance level shall be zero.

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4.3 Examination. The audiometer shall be examined to determine compliance with all the requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Any component part missing, broken or otherwise unusable.

102 Exposed edges not free of burs or sharpness.

103 Accessory storage compartment lining not suitably attached.

104 Any control dial or switch loose or does not operate smoothly.

105 Any indicator light does not operate properly.

106 Service data missing, not complete, or not as specified,

107 Headbands do not exert proper lateral pressure.

Minor

201 Identification marking not complete, permanent or legible.

202 Identification marking not securely attached to unit.

203 Finish not free of chipped areas, cracks, scratches, or roughness.

204 Any component part not securely held by mounts when not in use.

*Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests. Tests shall be conducted in accordance with ASA Standards, to determine compliance with this specification.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One audiometer, as specified, constitutes one unit of issue.

5.1.2 Level A. Component parts shall be suitably protected and secured in position within the carrying case in such a manner as to prevent shifting of or damage to the components. Each unit shall be packaged in accordance with MIL-P-116, Method II B, except that the outer container shall conform to PPP-B-636, type CF, class weather-resistant. Closure of carton shall be in accordance with the appendix of the box specification.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-601, table II, using type III, class 1 plywood or PPP-B-621, table III.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-621, table I.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked to identify contents.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Where to send first article, when required (see 3.1).

(c) When submission of first article is not required (see 3.1).

(d) When submission of drafts and extra copies of service data are not required (see 3.8).

(e) Where to send service data, when required (see 3.8).

(f) Selection of applicable levels of packing and packaging (see 5.1 and 5.2).

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Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

CATEGORY N

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

MIL-A-36494

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 60 1426 REPLACES EDITION OF 1 OCT 54, WHICH MAY BE USED.

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